GAUGE-FIXING FORESTS AND THE BAKRY–ÉMERY REGIME FOR LATTICE YANG–MILLS THEORY: AN INDEPENDENT DERIVATION

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Abstract. Shen, Zhu and Zhu showed that the Wilson lattice Yang–Mills measure with gauge group $\mathrm{SU}(N)$ and 't Hooft-scaled action $N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p$ satisfies the Bakry–Émery condition for $|\beta| < 1/(16(d-1))$, which yields a log-Sobolev inequality and a mass gap. The threshold comes from a bound $8(d-1)$ on the Hessian of the action. Fixing the gauge on a forest of links shortens plaquettes and lowers the plaquette-wise Cauchy–Schwarz constant to $c_F = \max_e \sum_{p\ni e} k_p$, where k_p is the number of free links in p. This idea, with a sharper constant $29/2$ in $d = 4$ and for all compact simple groups, and the sharp un-gauge-fixed bound $4d$ in every dimension, were obtained earlier in unrefereed preprints of Bhattacharjee, Bhattacharya and Lockwood. We give an independent derivation of the combinatorial part. We determine the minimum of c_F over all gauge-fixing forests: it is 18 in $d = 4$ and 11 in $d = 3$, with a kernel-checked Lean 4 certificate, and $8d-12$ for the temporal gauge; any improvement must therefore go beyond plaquette-wise Cauchy–Schwarz, as the cited work does. This gives $|\beta| < 1/36$ in $d = 4$ (weaker than the $1/29$ and $1/32$ claimed in the cited work) and $|\beta| < 1/22$ in $d = 3$, which, as far as we know, is not covered by those claims. We also give a short proof that the un-gauge-fixed condition fails for $|\beta| \geq 1/(8d)$, and independent numerical evidence for the sharp bound $4d$ and for the lower estimate $9 + \sqrt{17}$ on the gauge-fixed constant in $d = 4$. All results are at strong coupling and do not bear on the continuum limit.

1. Introduction

Let Λ be a finite subgraph of $\mathbb{Z}^d$, $d \geq 2$, with oriented edges E and positively oriented plaquettes P. For $G = \mathrm{SU}(N)$, $N \geq 2$, the Wilson lattice Yang–Mills measure in the 't Hooft scaling is

$$\mu_{\Lambda,\beta}(dQ) = Z^{-1} \exp\big(S(Q)\big) \prod_{e\in E} dQ_e, \qquad S(Q) = N\beta \sum_{p\in P} \mathrm{Re}\,\mathrm{Tr}\,Q_p, \tag{1}$$

with dQ_e normalised Haar measure and $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$ the plaquette holonomy. In standard lattice notation $\beta_{\mathrm{W}} = 2N/g^2 = N^2\beta$.

Shen, Zhu and Zhu [11] introduced a stochastic-analysis approach: they verify the Bakry–Émery condition [4]

$$\mathrm{Ric}(v,v) - \mathrm{Hess}\,S(v,v) \;\geq\; K\,|v|^2 \qquad \text{for all } Q \in G^E,\ v \in T_Q G^E, \tag{2}$$

with $K > 0$ uniform in Λ. With the metric $|X|^2 = \mathrm{Tr}(XX^*)$ on $\mathfrak{su}(N)$ the Ricci curvature of $\mathrm{SU}(N)$ is $\frac{N}{2}|X|^2$, and [11, Lemma 4.1] shows $|\,\mathrm{Hess}\,S(v,v)| \leq 8(d-1)N|\beta||v|^2$, so (2) holds with $K = \frac{N}{2} - 8(d-1)N|\beta|$ whenever

$$|\beta| < \frac{1}{16(d-1)}. \tag{3}$$

From (2) they derive a log-Sobolev inequality, exponential ergodicity of the Langevin dynamics, uniqueness of the infinite-volume measure, exponential decay of correlations (a mass gap) and large-N factorisation. The regime (3) improved on classical strong-coupling results [9]. Later

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work extended the method to Yang–Mills–Higgs [12], proved area law at $|\beta| < 1/(8(d-1))$ by applying it to the slab σ-models of Durhuus–Fröhlich [6, 5], treated U(N) [8], and proved a log-Sobolev inequality at all couplings in $d = 2$ [10].

Results. The constant $8(d-1)$ arises because each link lies in $2(d-1)$ plaquettes and each plaquette has four links: a plaquette with k links contributes k to the Hessian coefficient of each of its links. Our observation is that the four links need not all be free. Fixing the gauge on a forest $F \subset E$ (setting $Q_e = I$ for $e \in F$) is exact for gauge-invariant observables (Lemma 2.2) and replaces each plaquette by a loop with $k_p = 4 - |p \cap F|$ free links. The Hessian of the gauge-fixed action then obeys (Proposition 3.2)

$$|\operatorname{Hess} S^F(v,v)| \;\le\; c_F\, N|\beta|\,|v|^2, \qquad c_F = \max_{e\notin F} \sum_{p\ni e} k_p.$$

- The temporal gauge gives $c_F = 8d - 12$ in every dimension (Example 4.2).
- In $d = 4$ there is a period-2 forest in which every plaquette contains exactly one fixed link, so $c_F = 6 \cdot 3 = 18$; in $d = 3$ there is a period-3 forest with $c_F = 11$ (Theorem 4.3).
- These values are optimal among all gauge-fixing forests, *for the plaquette-wise Cauchy–Schwarz method*, even for the sharpest constant that method can give (the Perron eigenvalue of the free-edge incidence matrix): an integrality argument gives the lower bound 11 in $d = 3$ and 18 in $d = 4$ for every forest (Proposition 4.4). They are not lower bounds for the gauge-fixed Hessian itself.

Relation to prior work. After this work was done we found that the forest idea had appeared earlier, with stronger results, in unrefereed preprints by Bhattacharjee, Bhattacharya and Lockwood [3, 2], which we have not refereed in detail. They use a forest in $d = 4$ with exactly one fixed link per plaquette (as in our Theorem 4.3(a)), and bound the mixed second derivatives of each plaquette directly rather than by Cauchy–Schwarz over its free links. They claim $|\operatorname{Hess} S^F| \le \frac{29}{2} N|\beta|$, hence $|\beta| < 1/29$ for SU(N), together with extensions to every compact simple group, and they report that the optimal gauge-fixed constant lies between $9+\sqrt{17} \approx 13.12$ and $29/2$. A companion paper [1] claims the sharp un-gauge-fixed bound $|\operatorname{Hess} S| \le 4d\, N|\beta|$ in every dimension, i.e. our Conjecture 5.2, which gives $|\beta| < 1/(8d)$, and claims Dobrushin-type regimes beyond both. If these claims hold, our $d = 4$ result ($1/36$) is superseded both by their forest bound ($1/29$) and by the un-gauge-fixed $1/32$. What remains here is: an independent derivation of the forest construction; the optimality of 18 and 11 within plaquette-wise Cauchy–Schwarz, which explains why their finer block bound is needed; the $d = 3$ forest with $|\beta| < 1/22$, which improves on $1/24 = 1/(8 \cdot 3)$ and which we did not find in their work; a Lean certificate for the combinatorics; and independent numerics consistent with their claims (Section 5, and Remark 5.3).

Theorem 1.1. *Let $G = \mathrm{SU}(N)$, $N \ge 2$, and let F be one of the forests of Theorem 4.3. Let Λ be a box of $\mathbb{Z}^d$ with free boundary conditions. If*

$$|\beta| < \frac{1}{36} \;\; (d=4), \qquad |\beta| < \frac{1}{22} \;\; (d=3), \qquad |\beta| < \frac{1}{16d-24} \;\; (d \ge 2,\ \textit{temporal gauge}),$$

then the gauge-fixed measure $\mu^F_{\Lambda,\beta}$ satisfies the Bakry–Émery condition with $K = \frac{N}{2} - c_F N|\beta| > 0$, uniformly in Λ. In particular it satisfies a log-Sobolev and a Poincaré inequality with constant $1/K$, and for gauge-invariant $f \in L^2(\mu_{\Lambda,\beta})$

$$\mathrm{Var}_{\mu_{\Lambda,\beta}}(f) \le \frac{1}{K} \int \sum_{e\notin F} |\nabla_e \tilde f|^2 \, d\mu^F_{\Lambda,\beta}, \qquad \tilde f = f|_{Q_F = I}.$$

Moreover, the arguments of [11, Theorem 1.2, Corollary 1.6] *apply to μ^F and give exponential decay of correlations of local gauge-invariant observables, uniformly in Λ, and uniqueness of the infinite-volume limit of gauge-invariant expectations.*

For $\mathrm{SO}(N)$, $N \geq 3$, the same holds with $\frac{N}{2}$ replaced by $\frac{N-2}{4}$, i.e. for $|\beta| < \frac{N-2}{4Nc_F}$.

In standard normalisation, Theorem 1.1 covers $\beta_{\mathrm{W}} < N^2/36$ in $d = 4$, i.e. $\beta_{\mathrm{W}} < 0.111$ for $\mathrm{SU}(2)$ and $\beta_{\mathrm{W}} < 0.25$ for $\mathrm{SU}(3)$, against 0.083 and 0.1875 from (3); this is weaker than the $1/29$ claimed in [3]. In $d = 3$ the regime grows from $1/32$ to $1/22$.

Limits of the method. Section 5 records how far the Bakry–Émery route can go. Without gauge fixing, (2) fails for $|\beta| \geq 1/(8d)$: at a π-flux configuration (for $\beta > 0$) or the trivial one (for $\beta < 0$) the Hessian reaches $4d\,N|\beta|$ (Proposition 5.1). Adversarial numerical maximisation for $\mathrm{SU}(2)$ in $d = 2, 3$ approaches $4d$ from below, and we conjecture that $4d$ is the exact supremum on $\mathbb{Z}^d$ (Conjecture 5.2); a proof is claimed in [1]. The conjecture would give the regime $|\beta| < 1/(8d)$ without gauge fixing. In $d = 3$ the forest bound already does better ($1/22 > 1/24$); in $d = 4$ the conjecture would give $1/32$, somewhat beyond the proven $1/36$. In general the forest lower bound $2(d-1)\varphi(s_*)/s_* \geq 8(d-1)^2/d$ of Proposition 4.4 is below $4d$ only for $d \leq 3$.

What this is not. All statements concern the lattice theory at fixed lattice spacing and strong coupling. They do not address the continuum limit $\beta_{\mathrm{W}} \to \infty$ or the Clay problem [7]; and, in view of [3, 1], the improvement over [11] in $d = 4$ is not new.

2. Exact gauge fixing on a forest

A gauge transformation $g \in G^V$ acts by $(g \cdot Q)_{(x,y)} = g_x Q_{(x,y)} g_y^{-1}$; S and $\mu_{\Lambda,\beta}$ are invariant. A function f on G^E is gauge-invariant if $f(g \cdot Q) = f(Q)$ for all g.

Definition 2.1. For a forest $F \subset E$ (a set of edges containing no cycle) let $E' = E \setminus F$ and define on $G^{E'}$

$$\mu^F_{\Lambda,\beta}(dQ') = Z_F^{-1} \exp\left(S^F(Q')\right) \prod_{e \in E'} dQ'_e, \qquad S^F(Q') = S(I_F, Q').$$

Lemma 2.2. *For every gauge-invariant bounded measurable f, $\int f \, d\mu_{\Lambda,\beta} = \int f(I_F, \cdot)\, d\mu^F_{\Lambda,\beta}$.*

Proof. Root each component of F at a vertex r and set $g_r = I$; for an edge $(x, y) \in F$ with x closer to the root, set $g_y = g_x Q_{(x,y)}$ (and $g_y = g_x Q^{-1}_{(y,x)}$ for the opposite orientation). Vertices not met by F get $g = I$. Since F is a forest this is well defined, g depends only on Q_F, and $(g \cdot Q)_e = I$ for $e \in F$. By invariance, $\int f e^S dQ = \int f(g \cdot Q) e^{S(g \cdot Q)} dQ$. For fixed Q_F the map $Q_{E'} \mapsto (g \cdot Q)_{E'}$ is a product of left and right translations, which preserve Haar measure. Integrating first over $Q_{E'}$ and then over Q_F (a probability measure) gives $\int f(I_F, Q') e^{S^F(Q')} dQ'$. The same computation with $f = 1$ identifies the normalisations. □

Remark 2.3. On a torus the temporal links form closed lines, so the temporal gauge cannot be imposed everywhere. This is why Theorem 1.1 uses boxes with free boundary conditions (or $\mathbb{Z}^d$).

3. The Hessian of the gauge-fixed action

We follow [11]. Tangent vectors are $v = (X_e Q_e)_{e \in E'}$, where $X_e \in \mathfrak{su}(N)$, and $|v|^2 = \sum_e \mathrm{Tr}(X_e X_e^*)$; along the geodesics $t \mapsto e^{tX_e} Q_e$ one has $\mathrm{Hess}\, S^F(v, v) = \frac{d^2}{dt^2} S^F$.

Lemma 3.1. *Let ℓ be a loop whose holonomy is a product of k free link variables (each appearing once, possibly inverted) and fixed unitary matrices, and $\Phi = \operatorname{Re}\operatorname{Tr}(\text{holonomy})$. Then $|\operatorname{Hess}\Phi(v,v)| \le \big(\sum_{e\in\ell}|X_e|\big)^2 \le k\sum_{e\in\ell}|X_e|^2$.*

Proof. Differentiating twice inserts either one factor X_e^2 at the position of e, or two factors X_e, $X_{e'}$ at the positions of $e \neq e'$ (each unordered pair twice). Every resulting term has the form $\operatorname{Re}\operatorname{Tr}(AYBY')$ with A, B unitary and $Y, Y' \in \{\pm X_e, \pm X_{e'}\}$, so by Cauchy–Schwarz for the Hilbert–Schmidt inner product it is at most $|X_e||X_{e'}|$ in absolute value. Summing gives $(\sum_e |X_e|)^2$, and Cauchy–Schwarz gives the second inequality. □

Proposition 3.2. *For every $Q' \in G^{E'}$ and v,*

$$|\operatorname{Hess} S^F(v,v)| \le N|\beta| \sum_{e\in E'} c_e|X_e|^2 \le c_F N|\beta||v|^2,$$

where $c_e = \sum_{p\ni e} k_p$, $k_p = |p \setminus F|$ and $c_F = \max_{e\in E'} c_e$.

Proof. Apply Lemma 3.1 to each plaquette with $k = k_p$ and sum: $\sum_p k_p \sum_{e\in p\setminus F}|X_e|^2 = \sum_{e\in E'}|X_e|^2 \sum_{p\ni e} k_p$. □

Proof of Theorem 1.1. $G^{E'}$ is a product of copies of $\mathrm{SU}(N)$, so $\mathrm{Ric}(v,v) = \frac{N}{2}|v|^2$, and Proposition 3.2 gives (2) for S^F with $K = \frac{N}{2} - c_F N|\beta|$, which is positive under the stated conditions by Theorem 4.3 and Example 4.2. The log-Sobolev and Poincaré inequalities follow from the Bakry–Émery criterion [4]. The variance bound follows from Lemma 2.2 applied to f and f^2.

For the remaining statements we argue by adaptation rather than repeat the proofs: the proofs of [11, Theorem 1.2] (ergodicity and uniqueness) and [11, Corollary 4.11] (decay of correlations) use, besides (2) with uniform K, only that the action is a sum of finite-range terms with uniformly bounded derivatives, through the commutator estimate [11, Lemma 4.10] and the gradient bound [11, Lemma 4.9]. S^F is a sum of loops of length at most four with the same derivative bounds, so these proofs apply to μ^F with constants of the same form. (The Bakry–Émery condition, the log-Sobolev and Poincaré inequalities and the variance bound are proved in full above; uniqueness and decay of correlations are obtained by this adaptation.) For gauge-invariant local f, g, $\mathrm{Cov}_\mu(f,g) = \mathrm{Cov}_{\mu^F}(\tilde f, \tilde g)$ by Lemma 2.2, and $\tilde f$, $\tilde g$ are local. For $\mathrm{SO}(N)$ replace $\frac{N}{2}$ by $\frac{N-2}{4}$ [11]. □

4. Optimal forests

Definition 4.1. Label the coordinate directions $0, \dots, d-1$. For $\mu : \mathbb{Z}^d \to \{0, \dots, d-1\}$ let $F_\mu = \{(x - e_{\mu(x)}, x) : x \in \mathbb{Z}^d\}$.

The coordinate sum strictly decreases along $x \mapsto x - e_{\mu(x)}$, so F_μ contains no cycle: it is a forest for every μ, with exactly one fixed edge entering each vertex. Its restriction to a box is again a forest. The edge $(y, y+e_a)$ is fixed iff $\mu(y+e_a) = a$, and the plaquette $(y; a, b)$ contains

$$f_p = [\mu(y+e_a) = a] + [\mu(y+e_b) = b] + [\mu(y+e_a+e_b) \in \{a,b\}]$$

fixed edges. All quantities are periodic when μ is.

Example 4.2. $\mu \equiv 0$ (temporal gauge, with time direction 0): spatial plaquettes have $k_p = 4$, temporal ones $k_p = 2$; a spatial edge lies in $2(d-2)$ spatial and two temporal plaquettes, so $c_F = 8(d-2) + 4 = 8d - 12$.

Theorem 4.3. (a) *($d = 4$.) Let μ be 2-periodic, with values on $\{0,1\}^4$ (lexicographic order, first coordinate most significant, directions labelled $0, \dots, 3$) given by the string*

0212331001332120. Then every plaquette contains exactly one edge of F_μ, and $c_e = 18$ for every free edge.

(b) *($d = 3$.) Let μ be 3-periodic, with values on $\{0,1,2\}^3$ given by the string*

210102120102120210120210102.

Then $c_e \le 11$ for every free edge, with equality attained.

On a box with free boundary conditions, plaquettes outside the box are simply absent, so $c_F \le 18$, respectively ≤ 11.

Proof. Both statements are finite checks over one period cell. They were found by a CP-SAT search and verified by an independent script and by a kernel-checked Lean 4 computation (Appendix A). The Lean file also confirms that the temporal gauge has $c_e = 20$ in $d = 4$, and tamper controls (a changed pattern, or 11 replaced by 10) are rejected. □

Proposition 4.4. *Let $\varphi : [0,4] \to \mathbb{R}$ be the piecewise-linear interpolation of $k \mapsto k^2$ at the integers, and $s_* = 4(d-1)/d$. For every forest F on the box $\{0,\dots,L-1\}^d$, the matrix M_F on free edges defined by $M_F[e,e'] = \#\{p : e,e' \in p\}$ satisfies*

$$c_F \;\ge\; \rho(M_F) \;\ge\; 2(d-1)\frac{\varphi(s_*)}{s_*} - O(1/L).$$

The right-hand side equals 11 in $d = 3$ and 18 in $d = 4$. Since Lemma 3.1 summed over plaquettes gives $|\operatorname{Hess} S^F(v,v)| \le N|\beta|\langle |X|, M_F|X|\rangle \le \rho(M_F)N|\beta||v|^2$, and $\rho(M_F)$ is the best constant this inequality can give, the forests of Theorem 4.3 are optimal for this method in $d = 3$ and $d = 4$.

Proof. $c_F \ge \rho(M_F)$ because c_F is the maximal row sum of the nonnegative matrix M_F. Testing M_F against the constant vector gives $\rho(M_F) \ge \frac{1}{|E'|}\sum_{e\in E'} c_e = \frac{1}{|E'|}\sum_p k_p^2$, where $E' = E \setminus F$. Let $\bar k = \frac{1}{|P|}\sum_p k_p$. Since the k_p are integers and φ is the convex minorant of k^2 on integers, $\sum_p k_p^2 \ge |P|\varphi(\bar k)$. Moreover $\sum_p k_p = \sum_{e\in E'} n_e$, where $n_e = 2(d-1)$ except for $O(L^{d-1})$ boundary edges, so $|E'| = \frac{|P|\bar k}{2(d-1)} + O(L^{d-1})$ and $\rho(M_F) \ge 2(d-1)\varphi(\bar k)/\bar k - O(1/L)$. The function $\varphi(s)/s$ is nondecreasing on $[1,4]$ (on $[m,m+1]$ it equals $(2m+1) - m(m+1)/s$). Since F is a forest, $|E'| \ge |E| - |V| = (d-1)L^d - O(L^{d-1})$, and $|P| = \binom{d}{2}L^d + O(L^{d-1})$, so $\bar k \ge s_* - O(1/L)$. Finally $\varphi(8/3) = \frac{22}{3}$ gives $2\cdot 2\cdot \frac{22/3}{8/3} = 11$ for $d = 3$, and $\varphi(3) = 9$ gives $2\cdot 3\cdot 3 = 18$ for $d = 4$. □

Remark 4.5. Equality in Proposition 4.4 requires one fixed edge per vertex and every k_p to lie in $\{\lfloor s_*\rfloor, \lceil s_*\rceil\}$ in the right proportions. In $d = 4$ ($s_* = 3$) this forces exactly one fixed edge in every plaquette. In $d = 3$ ($s_* = 8/3$) it forces a $2:1$ mix of plaquettes with one and with two fixed edges. Both are realised by Theorem 4.3: on the 6^3 box, 300 plaquettes have one fixed edge and 150 have two.

5. Limits of the Bakry–Émery route

Proposition 5.1. *Let $G = \mathrm{SU}(N)$, $N \ge 2$, $\Lambda = \mathbb{Z}^d$ (or a torus of even side). Then for every $\beta \ne 0$, $\sup_{Q,v} \operatorname{Hess} S(v,v)/|v|^2 \ge 4d\,N|\beta|$. Consequently (2) without gauge fixing fails for every $K > 0$ when $|\beta| \ge 1/(8d)$.*

Proof. Let $J = \operatorname{diag}(-1,-1,1,\dots,1) \in \mathrm{SU}(N)$ and $Q_{(x,x+e_\mu)} = J^{x_0+\cdots+x_{\mu-1}}$. All links commute and every plaquette equals J. Take $X_e = \xi_e Z$ with $\xi \in \ell^2(E;\mathbb{R})$ and Z in the $\mathfrak{su}(2)$ block, so that $JZ = ZJ = -Z$ and $|Z| = 1$. All conjugations act trivially on Z, and the second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$ is $\operatorname{Re}\operatorname{Tr}\big((d\xi)_p^2 Z^2 J\big) = (d\xi)_p^2$, where d is the lattice coboundary. Hence

$\operatorname{Hess} S(v,v) = N\beta\|d\xi\|^2$, and $\sup_\xi \|d\xi\|^2/\|\xi\|^2 = 4d$ (Fourier mode $k = (\pi,\dots,\pi)$; on $\mathbb{Z}^d$ as a supremum). For $\beta < 0$ use $Q \equiv I$: then $\operatorname{Re}\operatorname{Tr}\big((d\xi)_p^2 Z^2\big) = -(d\xi)_p^2$ and $\operatorname{Hess} S(v,v) = N|\beta|\|d\xi\|^2$. Since $\operatorname{Ric}(v,v) = \frac{N}{2}|v|^2$ is constant, (2) with $K > 0$ forces $4d|\beta| < \frac{1}{2}$. □

Conjecture 5.2. For $G = \mathrm{SU}(N)$, $\Lambda = \mathbb{Z}^d$ and every configuration, $\operatorname{Hess} S(v,v) \le 4d\,N|\beta||v|^2$.

A proof of Conjecture 5.2, for $\mathrm{SU}(N)$, $\mathrm{SO}(N)$ and $\mathrm{U}(N)$ in every d, is claimed in [1]; we have not checked it, and we keep the statement as a conjecture here.

Evidence and a caution. For SU(2) we computed the exact Hessian, checked it against finite differences, and maximised $\lambda_{\max}$ over configurations by annealed single-link ascent. The starting points were Haar-random, centre-valued, abelian and perturbed π-flux configurations. On the tori $(L,d) = (2,2),(4,2),(2,3),(3,3),(2,4)$ every run stayed below $4d$, approaching it from below (best values 7.9999, 11.993 and 15.955 in $d = 2,3,4$). A shorter run on the $L = 3$, $d = 4$ torus also stayed below 16 but had not converged (best 13.1), so it is weak evidence. For abelian groups the bound is immediate, since there $\operatorname{Hess} S(v,v) = -N\beta\sum_p \cos\theta_p (d\xi)_p^2$; the difficulty in a proof is the commutator term. The naive covariant analogue of $\|d\|^2 \le 4d$ is false: for the covariant coboundary $(d_A X)_p = X_{x,\mu} + \operatorname{Ad}(Q_{x,\mu})X_{x+\mu,\nu} - X_{x,\nu} - \operatorname{Ad}(Q_{x,\nu})X_{x+\nu,\mu}$, ascent on the $L = 2$, $d = 4$ torus reaches $\|d_A\|^2 \approx 16.2 > 16$ (at that configuration the Hessian is only ≈ 8.4, and ascent started from it reaches 14.6). Any proof of Conjecture 5.2 therefore has to use the interplay between the commutator term and the trace weight $\operatorname{Re}\operatorname{Tr}(\,\cdot\, Q_p)$, not a covariant norm bound alone. We state the conjecture for $\mathbb{Z}^d$; on very small tori (where distinct plaquettes may share two edges) it may fail, although our searches on the $L = 2$ tori found no violation.

Conjecture 5.2 would give the un-gauge-fixed regime $|\beta| < 1/(8d)$. This is weaker than the forest bound in $d = 3$ and stronger in $d \ge 4$; in $d = 4$ it would be $1/32$ against our $1/36$.

Remark 5.3. For the gauge-fixed action in $d = 4$ the Cauchy–Schwarz constant 18 is not attained. At $Q \equiv I$, with the forest of Theorem 4.3(a) on L^4 boxes and SU(2), the largest eigenvalue of $-\operatorname{Hess} S^F/(N\beta)$ is 10.44, 11.54 and 12.11 for $L = 3,4,5$, increasing towards $9 + \sqrt{17} \approx 13.12$. This agrees with the lower estimate reported in [3].

6. Discussion

The mechanism behind the forest bound is the same one behind the factor-two gain of [5] for the slab σ-models: shorter interaction loops give smaller Cauchy–Schwarz constants. Here it is implemented on the full d-dimensional measure by a choice of gauge, and the best choice is a combinatorial design problem that can be solved exactly within the plaquette-wise Cauchy–Schwarz method; going further requires bounding each plaquette's mixed second derivatives as a block, as in [3]. The remaining gap between these thresholds and the weak-coupling regime relevant for the continuum limit ($\beta_{\mathrm{W}} \gtrsim 2.3$ for SU(2) in $d = 4$) is large, and closing it would require ideas of a different nature, such as a multiscale analysis.

Appendix A. Computer verification

The code is in the directory `earlier-note` of the public repository

`https://github.com/romanov360/yang-mills-3d-forest-gauge` :

`bakry_emery` holds the following files and `results` their outputs.

- `ForestCounting.lean`: Lean 4 (no Mathlib). It checks by `decide +kernel` that the $d = 4$ pattern has exactly one fixed edge in every plaquette and $c_e = 18$ on every free edge, that the $d = 3$ pattern has $c_e \le 11$, and that the temporal gauge has $c_e = 20$ in $d = 4$. The only axiom used is `propext`.

- `verify_forest.py`: an independent check. It confirms the forest property by union–find, the fixed-edge counts and c_e on 4^4, 6^4 and 6^3 boxes, and exactness of gauge fixing (all forest links become I and the action is unchanged). It also checks that the gauge-fixed SU(2) Hessian obeys the bound and agrees with finite differences.
- `forest_search.py`: the CP-SAT search. `perron.py`: the Perron eigenvalues $\rho(M_F)$ (11, 18, and 20 for the temporal gauge in $d = 4$).
- `hessian.py`, `fasthess.py`, `search2.py`: the exact Hessian and the adversarial search behind Conjecture 5.2.

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